

SPADEADAM, United Kingdom

WMO: 031650

Lat: 55.0503N

Lon: 2.5544W

Elev: 325

StdP: 97.48

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.9	-6.8	2.2	-4.1	-5.2	2.5	-2.2	16.8	7.6	15.1	6.3	3.0	260	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	22.1	16.5	20.1	15.3	18.8	14.7	17.5	20.8	16.4	19.1	15.5	17.6	4.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.9	18.2	15.2	11.2	17.0	14.8	10.9	16.5	50.3	21.2	47.1	18.8	44.2	17.9	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.4	11.1	-7.4	26.1	2.2	2.3	-9.0	27.7	-10.3	29.0	-11.5	30.3	-13.1	32.0		
(5)			-7.8	19.6	2.2	1.5	-9.4	20.6	-10.7	21.5	-11.9	22.3	-13.5	23.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	2.6	2.7	3.9	6.2	9.2	11.9	13.9	13.1	11.3	8.6	4.9	3.0	(6)
(7)		DBStd	4.90	2.90	2.84	2.74	2.80	3.00	2.55	2.45	1.85	2.35	2.70	2.99	3.46	(7)
(8)		HDD10.0	1247	230	203	191	118	52	9	0	1	13	58	155	217	(8)
(9)		HDD18.3	3912	488	437	448	363	284	194	140	163	212	302	404	475	(9)
(10)		CDD10.0	378	0	0	1	4	26	65	121	96	51	14	1	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(11)
(12)		CDH23.3	29	0	0	0	0	3	5	19	2	1	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	5.8	5.6	5.7	5.2	5.3	4.7	4.5	4.6	5.0	5.2	5.5	5.7	(14)
(15)	Precipitation	PrecAvg	1219	134	107	80	75	79	80	95	100	87	120	122	138	(15)
(16)		PrecMax	1549	234	273	152	145	139	197	168	219	142	227	308	362	(16)
(17)		PrecMin	856	12	22	23	30	19	19	35	18	10	29	56	32	(17)
(18)		PrecStd	189	57	68	31	30	36	43	37	48	34	53	56	70	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	11.1	14.0	19.0	22.2	23.8	26.0	21.1	21.0	16.2	13.0	11.2	(19)
(20)			MCWB	9.6	7.7	8.8	12.5	15.5	17.8	18.4	16.5	16.1	13.6	12.5	10.1	(20)
(21)		2%	DB	8.9	9.1	10.8	14.9	19.9	20.9	22.9	19.1	18.0	14.8	11.2	10.1	(21)
(22)			MCWB	8.7	8.1	8.4	10.3	13.9	15.4	17.2	15.4	14.8	13.2	10.6	9.4	(22)
(23)		5%	DB	7.9	8.1	9.0	12.8	16.9	18.9	20.9	17.9	16.2	13.8	10.2	9.1	(23)
(24)			MCWB	7.7	7.5	7.5	9.3	12.3	14.5	16.4	14.9	13.8	12.6	9.5	8.7	(24)
(25)		10%	DB	7.0	7.0	8.0	11.0	14.2	16.9	18.8	16.8	15.1	12.8	9.2	8.0	(25)
(26)			MCWB	6.7	6.5	7.1	8.6	10.8	13.2	15.2	14.4	13.2	11.7	8.7	7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.2	10.5	12.8	16.8	18.1	19.6	17.6	17.3	14.8	12.6	10.9	(27)
(28)			MCDWB	10.5	9.9	13.7	17.5	21.7	21.8	24.4	19.6	19.8	15.9	13.0	11.4	(28)
(29)		2%	WB	8.7	8.2	8.9	11.3	14.3	16.3	17.8	16.5	15.6	13.7	11.1	9.9	(29)
(30)			MCDWB	8.9	8.6	9.7	14.2	18.5	19.8	21.6	18.1	17.2	14.4	11.5	10.3	(30)
(31)		5%	WB	7.5	7.4	7.9	10.0	12.9	14.8	16.8	15.7	14.5	12.8	10.0	8.9	(31)
(32)			MCDWB	7.6	7.8	8.6	12.1	16.0	17.8	20.1	17.1	15.8	13.6	10.4	9.1	(32)
(33)		10%	WB	6.5	6.3	6.9	9.0	11.7	13.8	15.7	14.9	13.7	11.9	8.9	7.7	(33)
(34)			MCDWB	6.7	6.8	7.7	10.5	14.1	16.3	18.1	16.1	15.0	12.6	9.3	7.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.9	6.0	7.7	8.4	8.1	8.0	6.9	6.8	5.7	5.0	4.6	(35)
(36)			MCDBR	4.8	6.1	7.6	10.9	12.4	11.7	11.3	8.6	8.4	6.5	5.5	5.6	(36)
(37)			MCWBR	5.0	5.2	5.5	6.9	7.3	7.0	6.3	5.4	5.5	5.1	5.3	5.6	(37)
(38)		5% WB	MCDWR	4.8	5.3	6.6	9.3	10.9	10.2	9.9	7.1	7.1	5.8	5.4	5.6	(38)
(39)			MCWBR	5.0	5.0	5.4	6.5	6.8	6.7	6.1	5.3	5.4	5.3	5.5	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.322	0.351	0.371	0.372	0.371	0.371	0.361	0.344	0.330	0.305	0.292	(40)	
(41)		taud	2.277	2.369	2.315	2.286	2.324	2.350	2.351	2.393	2.447	2.445	2.353	2.249	(41)	
(42)		Ebn at Noon	615	759	817	851	871	874	866	854	821	742	621	542	(42)	
(43)		Edh at Noon	57	74	97	115	117	116	113	102	85	68	52	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.55	1.20	2.27	3.65	4.76	4.70	4.53	3.66	2.61	1.42	0.70	0.41	(44)	
(45)		RadStd	0.08	0.13	0.23	0.38	0.36	0.46	0.49	0.28	0.22	0.17	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page